

Statement of Purpose

A payment fraud incident involving a family acquaintance in my hometown made me curious about how fraudulent transactions could be identified before causing financial harm. She had lost ₹10,000 — money saved for her daughter's school fees. While exploring this problem, I began learning about fraud detection systems and built FraudLens. During optimisation, I spent several days reducing memory consumption during preprocessing after repeated crashes on my local machine. Debugging that error at 2 a.m. taught me a hard truth: machine learning isn't just about accurate models. It's about creating systems that work under real constraints, protecting trust in India's rapidly digitising economy.

I am a final-year B.Tech student in Computer Science and Engineering and have developed a strong interest in Machine Learning through building systems that address real-world challenges. My motivation came from seeing how ML enables intelligent decision-making beyond data analysis: identifying patterns, making predictions, and delivering measurable impact.

I began by learning core concepts — supervised learning, classification, regression, model evaluation, and feature engineering — through online courses and documentation. Rather than staying theoretical, I worked with real datasets and practical challenges. This taught me data preprocessing, handling missing values, addressing class imbalance, choosing evaluation metrics, and balancing model performance with efficiency. I learned that strong ML solutions require mastery of the full pipeline: from data preparation to deployment.

A defining milestone was building FraudLens, an AI-powered online payment fraud detection system for real-time identification of fraudulent transactions. Using the IEEE-CIS Fraud Detection dataset, I built an end-to-end pipeline. While experimenting with multiple models, I found that reducing the feature space significantly improved efficiency without sacrificing accuracy. I reduced 394 engineered features to 53 optimised features, using PCA for dimensionality reduction and SHAP for feature selection. The final model achieved a mean AUC-ROC of 0.9756 via cross-validation. I also added K-Means clustering for pattern discovery and deployed the system as an interactive Streamlit application for real-time predictions.

The project changed my perspective on machine learning. I initially viewed success as achieving higher accuracy scores, but I learned that reliability, scalability, and interpretability are equally important when models affect real users.

I presented FraudLens at the national-level IdeaCode Hackathon, where my team ranked among the Top 3. Presenting the project exposed me to questions about latency and scalability that I had not considered during model development. I was the primary ML engineer: I led feature engineering, model selection, and

deployment. Mentors questioned inference latency under load, prompting me to optimise memory and batch predictions. Solving these issues — handling unseen categorical values, memory optimisation, and deployment reliability — deepened my understanding of scalable, production-ready ML systems. The project showed me how ML can address critical financial security challenges affecting millions of transactions.

Through FraudLens, I learned how model accuracy alone is insufficient for real-world deployment. I am applying to the Amazon ML Summer School because I'm particularly interested in understanding how large-scale ML systems balance accuracy, latency, interpretability, and reliability — challenges that become critical when models operate on millions of transactions. I want to deepen my understanding of deep learning, generative AI, optimization techniques, and distributed training, especially model interpretability at scale. Learning from Amazon researchers will help me bridge academic knowledge and real-world applications.

My long-term goal is to develop machine learning systems that improve trust and security in digital financial ecosystems, particularly for rapidly growing digital payment platforms. I believe Amazon ML Summer School will strengthen my technical foundation, broaden my understanding of modern ML research, and accelerate my growth as a practitioner. I'm excited to learn from experts, collaborate with motivated peers, and apply these learnings to develop impactful ML solutions.